

## Compound risk models

▮ The counting / frequency component

$N \in \{0, 1, 2, \dots\}$  models

- the # of compromises in a cyber system;
- the # of claims in an insurance contract ;
- the # of catastrophes in a given area.



▮ The measurement / severity component

$X_i > 0$  represents

- the financial consequence due to the  $i$ -th compromised device ;
- the \$ of payment of the  $i$ -th insurance claim ;
- the \$ of loss due to the occurrence of the  $i$ -th catastrophe.



III The aggregate risk / loss

$$S = \sum_{i=1}^N X_i, \text{ with } \sum_{i=1}^0 X_i = 0.$$

III Some common assumptions :

- $N$  and  $X_i$ 's are independent

- $X_i$ 's are independently and identically distributed (i.i.d.) with

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2.$$

- $E(N) = \lambda$  and  $\text{Var}(N) = b^2$



▮ If  $N = n$ , then

$$E(S|N=n) = E(X_1 + \dots + X_n)$$

$$= \underbrace{E(X_1) + \dots + E(X_n)}_n = n \cdot \mu$$

▮ If the value of  $N$  is unspecified, then

$$E(S|N) = N \cdot \mu$$

$$\text{▮ } E(S) = E(E(S|N))$$

$$= E(N \cdot \mu) = \mu \cdot E(N)$$

$$= \mu \cdot \lambda$$



$$\text{III} \quad \text{Var}(S) = E[\underbrace{\text{Var}(S|N)}_{\rightarrow \text{Var}(X_1 + \dots + X_n)}] + \text{Var}(E(S|N))$$

$$= E\left[ N \cdot \cancel{\text{Var}(X_1)}^{\theta^2} \right] + \text{Var}(N \cdot \mu)$$

$$= \theta^2 \cdot E(N) + \mu^2 \cdot \text{Var}(N)$$

$$= \theta^2 \cdot \lambda + \mu^2 \cdot b^2$$